FUNDAMENTALS OF DATA SCIENCE

T. Niranjan Kumar
192473007

Q1. Calculate the customer's z-score.

Answer

The customer's **Z-score is 3**.

Given

- Mean (μ) = ₹2000
- Standard Deviation (σ) = ₹400
- Customer Spending (X) = ₹3200

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Calculation

$$Z = \frac{3200 - 2000}{400}$$
$$= \frac{1200}{400}$$
$$= 3$$

Steps

1. Write the given values.
2. Apply the Z-score formula.
3. Substitute the values into the formula.
4. Perform the calculation.
5. Write the final answer.

Final Answer

Customer's Z-score = 3

Q2. Is the customer's spending unusually high compared with the average?

Answer

Yes. The customer's spending is unusually high because the Z-score is **3**.

Interpretation Table

Z-score	Interpretation
$Z > 2$	Unusually High
$-2 \leq Z \leq 2$	Normal
$Z < -2$	Unusually Low

Steps

1. Take the calculated Z-score.
2. Compare it with the interpretation table.
3. Since **3 > 2**, the spending is unusually high.
4. Write the conclusion.

Final Answer

The customer's spending is unusually high.

Q3. What would a negative z-score mean in this context?

Answer

A negative Z-score indicates that the customer spends **below the average amount**.

Interpretation Table

Z-score	Meaning
Positive (+)	Above Average
Zero (0)	Equal to Average
Negative (-)	Below Average

Steps

1. Observe the sign of the Z-score.
2. A negative sign means the value is below the mean.
3. Therefore, the customer spends less than the average.

Final Answer

A negative Z-score means below-average spending.

Q4. If another customer has a z-score of −1.5, what does it tell you?

Answer

The customer spends **1.5 standard deviations below the average spending**.

Interpretation Table

Customer	Z-score	Interpretation
Customer 1	+3	Unusually High Spending
Customer 2	-1.5	Below Average Spending

Steps

1. Note the given Z-score (−1.5).
2. A negative Z-score indicates below-average spending.
3. The value **1.5** shows the customer is **1.5 standard deviations below the mean**.
4. Write the conclusion.

Final Answer

The customer spends 1.5 standard deviations below the average.

CASE STUDY – 7: Delivery Time Analysis

Given Data

Delivery Time (minutes)

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Q1. Calculate the Mean and Median.

Answer

- **Mean = 32.6 minutes**
- **Median = 30 minutes**

Given

Delivery Time = 25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Number of observations (**n**) = 10

Formula

Mean

$$\text{Mean} = \frac{\sum X}{n}$$

Calculation

$$\sum X = 25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60 = 326$$

$$\text{Mean} = \frac{326}{10} = 32.6$$

Median

Since **n = 10 (even number)**,

$$\text{Median} = \frac{5^{th} + 6^{th} \text{ values}}{2}$$

$$= \frac{30 + 30}{2}$$

$$= 30$$

Steps

1. Find the total of all observations.
2. Divide the total by the number of observations to get the mean.
3. Arrange the data in ascending order.
4. Find the middle two values.
5. Calculate their average to get the median.

Final Answer

- **Mean = 32.6 minutes**
- **Median = 30 minutes**

Q2. Find Q1 and Q3.

Answer

- **Q1 = 28**
- **Q3 = 32**

Given

Data (Ascending Order):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Calculation

Lower Half:

25, 27, 28, 29, 30

Q1 = 28

Upper Half:

30, 31, 32, 34, 60

Q3 = 32

Steps

1. Arrange the data in ascending order.
2. Divide the data into two halves.
3. Find the median of the lower half (Q1).
4. Find the median of the upper half (Q3).

Final Answer

- **Q1 = 28**
- **Q3 = 32**

Q3. Calculate the IQR.

Answer

IQR = 4

Given

- Q1 = 28
- Q3 = 32

Formula

$$IQR = Q3 - Q1$$

Calculation

$$IQR = 32 - 28 = 4$$

Steps

1. Write Q1 and Q3.
2. Apply the IQR formula.
3. Subtract Q1 from Q3.
4. Write the result.

Final Answer

IQR = 4

Q4. Check for Outliers using the 1.5 × IQR Rule.

Answer

60 is an outlier.

Given

- Q1 = 28
- Q3 = 32

- IQR = 4

Formula

Lower Limit

$$Q1 - 1.5(IQR)$$

Upper Limit

$$Q3 + 1.5(IQR)$$

Calculation

Lower Limit

$$28 - (1.5 \times 4)$$
$$28 - 6 = 22$$

Upper Limit

$$32 + (1.5 \times 4)$$
$$32 + 6 = 38$$

Since **60 > 38**, it is an outlier.

Steps

1. Calculate the IQR.
2. Find the lower limit.
3. Find the upper limit.
4. Compare all observations with these limits.
5. Identify the outlier.

Final Answer

60 is an Outlier.

Q5. Explain whether the Mean or Median gives a better description of the typical delivery time.

Answer

Median gives a better description of the typical delivery time because the dataset contains an outlier (60), which increases the mean. The median is not affected by outliers and better represents the central value.

Comparison Table

Measure	Value	Remark
Mean	32.6	Affected by Outlier
Median	30	Better Represents Typical Delivery Time

Steps

1. Compare the mean and median.
2. Observe the presence of an outlier.
3. Understand that the mean is affected by extreme values.
4. Conclude that the median is a better measure.

Final Answer

Median is the better measure of the typical delivery time because it is not affected by the outlier.

CASE STUDY – 8: Two Cities' Temperatures

Given Data

City A (°C) 28 29 30 30 31 30 32

City B (°C) 20 25 30 35 30 25 45

Q1. Calculate the mean temperature for each city.

Answer

- **Mean of City A = 30°C**
- **Mean of City B = 30°C**

Given

City A = 28, 29, 30, 30, 31, 30, 32

City B = 20, 25, 30, 35, 30, 25, 45

Number of observations (**n**) = 7

Formula

$$\text{Mean} = \frac{\sum X}{n}$$

Calculation

City A

$$28 + 29 + 30 + 30 + 31 + 30 + 32 = 210$$
$$\text{Mean} = \frac{210}{7} = 30^{\circ}C$$

City B

$$20 + 25 + 30 + 35 + 30 + 25 + 45 = 210$$
$$\text{Mean} = \frac{210}{7} = 30^{\circ}C$$

Steps

1. Find the sum of all temperatures.
2. Count the number of observations.
3. Apply the mean formula.

4. Calculate the mean.
5. Write the final answer.

Final Answer

- **City A Mean = 30°C**
- **City B Mean = 30°C**

Q2. Calculate the variance for each city.

Answer

- **Variance of City A ≈ 1.43**
- **Variance of City B ≈ 57.14**

Given

Mean of both cities = **30°C**

Formula

$$\text{Variance} = \frac{\sum(X-\mu)^2}{N}$$

Calculation

City A

X	X − 30	(X − 30)²
28	-2	4
29	-1	1
30	0	0
30	0	0
31	1	1
30	0	0
32	2	4

$$\sum(X-\mu)^2 = 10$$
$$Variance = \frac{10}{7} = 1.43$$

City B

X	X − 30	(X − 30)²
20	-10	100
25	-5	25
30	0	0
35	5	25
30	0	0
25	-5	25
45	15	225

$$\sum (X - \mu)^2 = 400$$

$$Variance = \frac{400}{7} = 57.14$$

Steps

1. Find the mean.
2. Calculate deviation (X−Mean).
3. Square each deviation.
4. Find the total.
5. Divide by N.

Final Answer

- **City A Variance = 1.43**
- **City B Variance = 57.14**

Q3. Which city has greater temperature variability?

Answer

City B has greater temperature variability.

Comparison Table

City	Variance	Variability
City A	1.43	Low
City B	57.14	High

Steps

1. Compare both variances.
2. Higher variance means higher variability.
3. City B has the larger variance.
4. Therefore, City B has greater variability.

Final Answer

City B has greater temperature variability.

Q4. Why can two datasets have the same mean but different standard deviations?

Answer

Two datasets can have the **same mean** but **different standard deviations** because the values may be spread differently around the mean. One dataset may have values close to the mean, while the other has values far from the mean.

Comparison Table

Dataset	Mean	Spread
City A	30	Low Spread
City B	30	High Spread

Steps

1. Compare the means.
2. Observe the spread of values.
3. Greater spread results in a higher standard deviation.
4. Hence, both datasets can have the same mean but different standard deviations.

Final Answer

Because the spread of values around the mean is different.

Q5. Which city has more predictable temperatures?

Answer

City A has more predictable temperatures.

Comparison Table

City	**Predictability**
City A	More Predictable
City B	Less Predictable

Steps

1. Compare the variances.
2. Lower variance means more consistency.
3. More consistency means better predictability.
4. Therefore, City A is more predictable.

Final Answer

City A has more predictable temperatures.

CASE STUDY – 9: Data Center CPU Usage

Given Data

CPU Usage (%)

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Q1. Calculate the Mean, Median and Mode.

Answer

- **Mean = 51.3%**
- **Median = 47%**
- **Mode = 45%**

Given

CPU Usage = 40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Number of observations (**n**) = 10

Formula

Mean

$$\text{Mean} = \frac{\sum X}{n}$$

Calculation

$$40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90 = 513$$
$$\text{Mean} = \frac{513}{10} = 51.3$$

Median

Since **n = 10 (Even)**

$$\text{Median} = \frac{46 + 48}{2}$$
$$= \frac{94}{2} = 47$$

Mode

45 appears **2 times**, while all other values appear only once.

Therefore,

Mode = 45

Steps

1. Find the total of all observations.
2. Divide by the number of observations to get the mean.
3. Arrange the data in ascending order.
4. Find the middle two values and calculate the median.
5. Identify the value that occurs most frequently to find the mode.

Final Answer

- **Mean = 51.3%**
- **Median = 47%**
- **Mode = 45%**

Q2. Calculate the Range.

Answer

Range = 50

Given

- Maximum Value = 90
- Minimum Value = 40

Formula

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Calculation

$$90 - 40 = 50$$

Steps

1. Identify the maximum value.
2. Identify the minimum value.

3. Subtract the minimum from the maximum.
4. Write the final answer.

Final Answer

Range = 50

Q3. Find Q1, Q3 and IQR.

Answer

- **Q1 = 45**
- **Q3 = 52**
- **IQR = 7**

Calculation

Lower Half:

40, 42, 45, 45, 46

Q1 = 45

Upper Half:

48, 50, 52, 55, 90

Q3 = 52

Formula

$$IQR = Q3 - Q1$$

Calculation

$$52 - 45 = 7$$

Steps

1. Arrange the data in ascending order.
2. Divide the data into two halves.
3. Find the median of the lower half (Q1).
4. Find the median of the upper half (Q3).
5. Calculate IQR.

Final Answer

- **Q1 = 45**
- **Q3 = 52**
- **IQR = 7**

Q4. Identify Possible Outliers.

Answer

90 is a possible outlier.

Given

- Q1 = 45
- Q3 = 52
- IQR = 7

Formula

Lower Limit

$$Q1 - 1.5(IQR)$$

Upper Limit

$$Q3 + 1.5(IQR)$$

Calculation

Lower Limit

$$45 - (1.5 \times 7)$$
$$45 - 10.5 = 34.5$$

Upper Limit

$$52 + (1.5 \times 7)$$
$$52 + 10.5 = 62.5$$

Since **90 > 62.5**, it is a possible outlier.

Steps

1. Find the IQR.
2. Calculate the lower limit.
3. Calculate the upper limit.
4. Compare all observations with the limits.
5. Identify the outlier.

Final Answer

90 is a Possible Outlier.

Q5. Determine the Direction of Skewness.

Answer

The dataset is **Right-Skewed (Positively Skewed).**

Comparison Table

Observation	**Interpretation**
Mean (51.3) > Median (47)	Right Skew
High Value (90)	Pulls the Mean to the Right

Steps

1. Compare the mean and median.
2. Observe the presence of the extreme value (90).
3. Since the mean is greater than the median, the distribution is right-skewed.
4. Write the conclusion.

Final Answer

The CPU usage data is Right-Skewed (Positively Skewed).

CASE STUDY – 10: Comparing Two Datasets

Given Data

Dataset A 10 20 30 40 50

Dataset B 28 29 30 31 32

Q1. Calculate the Mean of both datasets.

Answer

- **Mean of Dataset A = 30**
- **Mean of Dataset B = 30**

Given

Dataset A = 10, 20, 30, 40, 50

Dataset B = 28, 29, 30, 31, 32

Number of observations (**n**) = 5

Formula

$$\text{Mean} = \frac{\sum X}{n}$$

Calculation

Dataset A

$$10 + 20 + 30 + 40 + 50 = 150$$
$$\text{Mean} = \frac{150}{5} = 30$$

Dataset B

$$28 + 29 + 30 + 31 + 32 = 150$$
$$\text{Mean} = \frac{150}{5} = 30$$

Steps

1. Find the sum of all values.
2. Count the number of observations.
3. Apply the mean formula.

4. Calculate the mean.
5. Write the final answer.

Final Answer

- **Dataset A Mean = 30**
- **Dataset B Mean = 30**

Q2. Calculate the Variance of both datasets.

Answer

- **Variance of Dataset A = 200**
- **Variance of Dataset B = 2**

Formula

$$\text{Variance} = \frac{\sum(X-\mu)^2}{N}$$

Calculation

Dataset A (Mean = 30)

X	X − 30	(X − 30)²
10	-20	400
20	-10	100
30	0	0
40	10	100
50	20	400

$$\sum(X-\mu)^2 = 1000$$

$$Variance = \frac{1000}{5} = 200$$

Dataset B (Mean = 30)

X	X − 30	(X − 30)²
28	-2	4
29	-1	1
30	0	0
31	1	1
32	2	4

$$\sum(X-\mu)^2 = 10$$

$$Variance = \frac{10}{5} = 2$$

Steps

1. Find the mean.
2. Calculate each deviation.
3. Square the deviations.
4. Find the total.
5. Divide by N.

Final Answer

- **Dataset A Variance = 200**
- **Dataset B Variance = 2**

Q3. Calculate the Standard Deviation of both datasets.

Answer

- **Dataset A Standard Deviation = 14.14**
- **Dataset B Standard Deviation = 1.41**

Formula

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

Calculation

Dataset A

$$SD = \sqrt{200} = 14.14$$

Dataset B

$$SD = \sqrt{2} = 1.41$$

Steps

1. Find the variance.
2. Apply the standard deviation formula.
3. Calculate the square root.
4. Write the final answer.

Final Answer

- **Dataset A SD = 14.14**
- **Dataset B SD = 1.41**

Q4. Which dataset has greater spread?

Answer

Dataset A has greater spread because it has a higher variance and standard deviation.

Comparison Table

Dataset	Variance	Standard Deviation	Spread
Dataset A	200	14.14	Greater
Dataset B	2	1.41	Smaller

Steps

1. Compare the variances.
2. Compare the standard deviations.
3. The dataset with the higher values has greater spread.
4. Write the conclusion.

Final Answer

Dataset A has greater spread.

Q5. If both datasets represent model errors, which model appears more consistent?

Answer

Dataset B appears more consistent because it has a much lower variance and standard deviation.

Comparison Table

Dataset	Consistency
Dataset A	Less Consistent
Dataset B	More Consistent

Steps

1. Compare the standard deviations.
2. Lower standard deviation indicates higher consistency.
3. Dataset B has the lower standard deviation.
4. Therefore, Dataset B is more consistent.

Final Answer

Dataset B is more consistent.

Q6. Explain why mean alone is not enough to compare datasets.

Answer

The **mean alone is not enough** because two datasets can have the same mean but different spreads. Variance and standard deviation show how the values are distributed around the mean, making comparison more accurate.

Comparison Table

Measure	Dataset A	Dataset B
Mean	30	30
Variance	200	2
Standard Deviation	14.14	1.41

Steps

1. Compare the means.
2. Observe that both means are equal.
3. Compare the variance and standard deviation.

4. Conclude that spread is also important.

Final Answer

Mean alone cannot compare datasets completely because it does not show the variation in data. Variance and standard deviation must also be considered.